

Data Cleaning & Structural Validation, you can create a simple project in Python

Project Structure

```
data-cleaning-project/  
|  
├── raw_data.csv  
├── cleaned_data.csv  
├── data_cleaning.py  
└── README.md
```

Sample Dataset (raw_data.csv)

```
Csv  
Name,Age,Email  
Kumar,20,kumar@gmail.com  
Kumar,20,kumar@gmail.com  
Priya,,priya@gmail.com  
Rahul,22,rahulgmail.com  
Anu,21,anu@gmail.com
```

Python Code (data_cleaning.py)

```
import pandas as pd  
  
# Load dataset  
df = pd.read_csv("raw_data.csv")  
  
# Remove duplicate rows  
df = df.drop_duplicates()  
  
# Fill missing age with average age  
df["Age"] = df["Age"].fillna(df["Age"].mean())  
  
# Remove invalid emails  
df = df[df["Email"].str.contains("@", na=False)]  
  
# Save cleaned data  
df.to_csv("cleaned_data.csv", index=False)  
  
print("Data cleaning completed successfully!")  
Python
```

README.md

Markdown

Data Cleaning & Structural Validation

Objective

Clean raw data by:

- Removing duplicates
- Handling missing values
- Validating email formats
- Producing a clean dataset

Tools Used

- Python
- Pandas

Output

A cleaned CSV file ready for analysis.